

Resistive-Probe Observations

Decide what copied evidence supports—and what it cannot

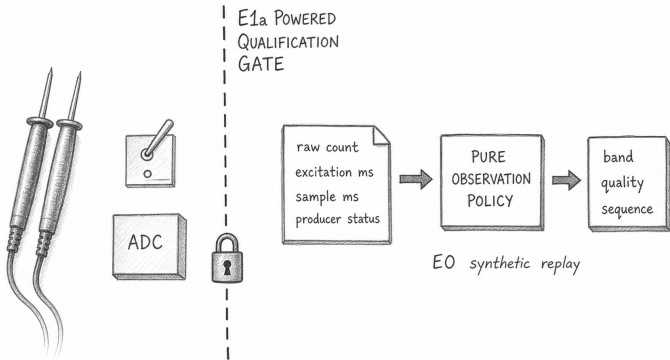
E0 SYNTHETIC REPLAY — DO NOT WIRE OR ENERGIZE A PROBE

Lesson	061 — component	Time	150–210 minutes
Prerequisites	status, supplied time, copied evidence	Board	compile-only Mega replay
Input	one complete acquisition-cycle sample	Observation	named memory result cell

Responsibility. `ResistiveProbeObservationPolicy` qualifies one copied conductive-probe sample against fixed dry and wet references, explicit excitation evidence, freshness, and a bounded per-cycle duty contract. It does not switch power, read an ADC, identify a liquid, estimate depth, compensate contamination, or prove that a physical leak is absent.

1 Start with an honest evidence boundary

A conductive probe can produce an ADC-like number, but the number alone does not explain the circuit that produced it. Geometry, ionic content, contamination, corrosion, supply conditions, and acquisition timing can all change the result. Lesson 061 therefore begins after acquisition: a synthetic producer supplies a complete copied record, and a pure policy decides which bounded interpretation that record supports.



Alternative text: a pencil-drawn boundary separates an unqualified physical probe, excitation switch, and ADC on the left from a copied sample card and pure observation policy on the right; a locked E1a gate blocks every wire and powered claim.

E0 proves	E0 does not prove
deterministic classification of copied synthetic samples	an ADC channel or excitation pin was acquired
fixed calibration and threshold arithmetic	a kit probe has a particular topology or polarity
explicit off, timing, age, and provenance checks	excitation physically turned off
bounded volatile policy state	corrosion, current, spill, and drying safety

There is no formal electrical schematic in this lesson. An authoritative schematic requires one exact specimen, its switched-excitation circuit, ADC reference and source-impedance evidence, and measured safe-state behavior. Those are E1a tasks, not blanks to fill with a generic module drawing.

2 Copy one complete acquisition cycle

The producer supplies the energized reading and the discharged reading from the same named acquisition cycle. Identity and order travel with those numbers: nonzero source, configuration, and calibration revisions; nonzero producer sequence; observation time; producer status; on-time; cycle time; and an explicit statement that excitation was observed off after sampling.

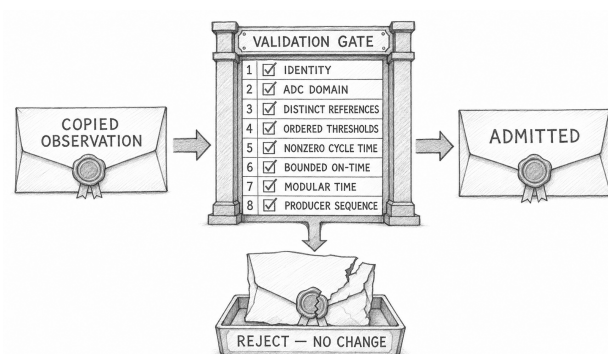
Alternative text: a hand-drawn sample card groups provenance, energized and discharged readings, excitation on-time and cycle time, observed off evidence, producer status, and observation time as one copied record.

Field group	Question it makes answerable
source and revisions	Which fixed producer and interpretation created this record?
sequence and time	Is it ordered, unambiguous, and fresh?
energized and discharged raw values	What happened during sampling and after excitation?
on-time and cycle time	What duty did this supplied cycle report?
off evidence and status	Did the producer observe cleanup, and did it report failure?

The policy copies the complete input and retains no caller pointer. Equal timestamps are idempotent only for byte-identical duplicates. A changed duplicate, backward time, exact modular half-range ambiguity, sequence regression, or changed source/revision rejects atomically.

3 Validate structure before interpreting liquid

Configuration fixes the declared ADC maximum, dry and wet references, disconnected and discharged limits, ordered damp and wet thresholds, maximum age, maximum on-time, and maximum duty. The ADC maximum must be nonzero. All raw values, calibration references, and disconnected and discharged limits must fit its declared domain. Dry and wet references must differ, while either slope is allowed.



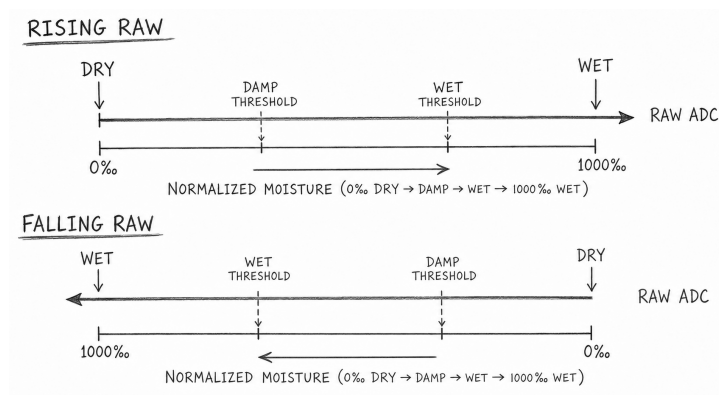
Alternative text: pencil checkpoints inspect identity, ADC domain, distinct references, ordered thresholds, nonzero cycle time, bounded on-time, modular time, and producer sequence before a sealed observation card is allowed to change.

1. Check lifecycle, fixed configuration, source identity, revisions, sequence, and modular time.
2. Check raw values and references against the declared ADC domain.
3. Require nonzero cycle time and on-time no greater than cycle time.
4. Compute duty with widened arithmetic so $\text{onTime} * 1000 / \text{cycleTime}$ cannot overflow.
5. Only after structural admission, apply the quality precedence.

Predict the result when a malformed cycle also contains a convincingly wet energized value: _____. Explain why interpretation cannot precede structure: _____.

4 Normalize either calibration slope

Normalization maps the dry reference to 0 permille and the wet reference to 1000 permille. Values between them are monotonic. Values beyond either reference clamp to the nearest endpoint for ordinary classification. The arithmetic must work whether wet is numerically above or below dry.



Alternative text: two pencil number lines show rising and falling calibrations; both map their dry reference to zero permille and wet reference to one thousand, with the damp and wet thresholds crossed in the same normalized order.

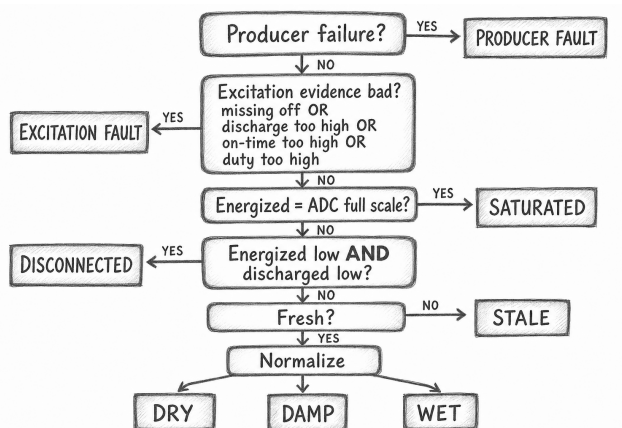
Normalized value	Threshold relationship	Ordinary quality
below damp threshold	dry side of both thresholds	Dry
at or above damp, below wet	between ordered thresholds	Damp
at or above wet	wet side of both thresholds	Wet

Fill one rising calibration with dry = 200, wet = 800. What normalized value should raw = 500 produce? _____ permille. Reverse the references. What should raw = 500 produce now? _____ permille.

Threshold tests inspect immediately below, exactly at, and immediately above each boundary. Integer arithmetic defines these boundaries; a replay must not depend on floating-point rounding.

5 Give faults precedence over an attractive number

A valid update applies a fixed order. Producer failure precedes every domain quality. Missing off evidence, excessive discharged reading, excessive on-time, or excessive duty produces **ExcitationFault**. Exact ADC full-scale energized input produces **Saturated**. The deliberately narrow disconnected predicate requires *both* energized and discharged values at or below the disconnected maximum.



Alternative text: a pencil decision path checks producer failure, excitation and discharge evidence, exact full-scale saturation, the two-low-values disconnected predicate, freshness, and only then normalized dry, damp, or wet classification.

Evidence	Why it dominates	Quality
producer reports failure	sample meaning is not trustworthy	ProducerFault
off missing or discharge/on-time/duty excessive	cleanup contract failed	ExcitationFault
energized raw equals ADC maximum	rail cannot be relabeled “very wet”	Saturated
both raw values at or below disconnected maximum	exact qualified synthetic predicate	Disconnected
age exceeds maximum	otherwise valid evidence arrived too late	Stale
all earlier checks pass	calibrated thresholds may be interpreted	dry/damp/wet

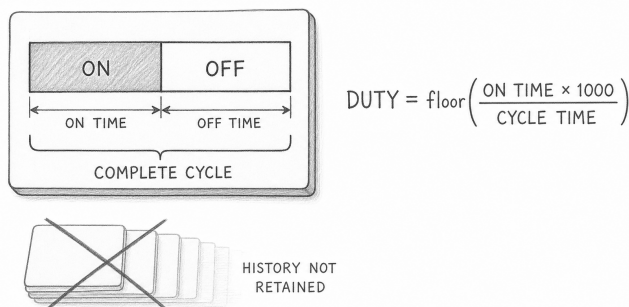
An arbitrary open circuit need not read low, and arbitrary contamination need not resemble the calibration fixture. The disconnected rule is a fixed synthetic contract awaiting E1 qualification, not a universal electrical diagnosis. Likewise, **Dry** means only “classified on the dry side of this fixed calibration”; it never certifies that a case has no leak.

6 Measure only the supplied cycle’s duty

For one admitted sample, observed cycle duty is

$$\left\lfloor \frac{\text{excitation on-time} \times 1000}{\text{cycle time}} \right\rfloor.$$

The policy checks this value and the independent maximum on-time. It retains no rolling energy history, so it cannot prove cumulative energized time, long-term corrosion duty, or compliance between supplied samples.



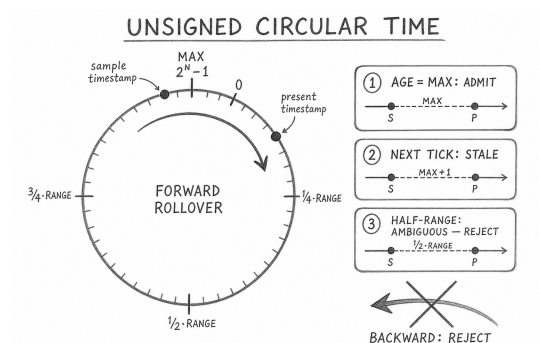
Alternative text: a pencil timing strip shows one bounded cycle with a short shaded excitation interval and a longer discharged interval; a bracket labels per-cycle duty while a broken line beyond the cycle is marked unknown history.

On-time	Cycle time	Duty	Interpretation
0	100 ms	0 permille	no reported excitation in this cycle
5 ms	100 ms	50 permille	supplied-cycle result only
100 ms	100 ms	1000 permille	continuously on for this cycle
101 ms	100 ms	invalid	structural rejection before division

For 7 ms on during a 64 ms cycle, predict the truncated duty: _____ permille. Then name the evidence still needed to make a physical corrosion-duty claim: _____.

7 Make age and ordering deterministic

TimePoint now is the only clock. Age is derived with modular unsigned time rules; every configured duration remains strictly below the half range. Evidence at exactly the maximum age is admitted. Evidence one tick older is **Stale**. Backward time and exact half-range ambiguity reject without changing the last accepted observation.



Alternative text: a hand-drawn circular counter crosses unsigned rollover from a late observation to a small current time; tick marks show the inclusive maximum-age boundary, the first stale tick, and a forbidden half-range chord.

Replay these cases before running tests:

1. an otherwise-dry observation at the inclusive age limit: stale or admitted? _____;
2. same timestamp and sequence with byte-identical content: disposition _____;
3. same timestamp and sequence with one changed raw value: disposition _____;
4. sequence `0xFFFFFFFF` followed by `0xFFFFFFFF` across ordinary clock rollover: accepted or rejected? _____;
5. after accepting `0xFFFFFFFF`, attempt a duplicate and then sequence 1: what status must both return? _____;
6. after explicit reset, attempt sequence 1: accepted or rejected? _____;
7. an exact half-range time difference: accepted or rejected as ambiguous? _____.

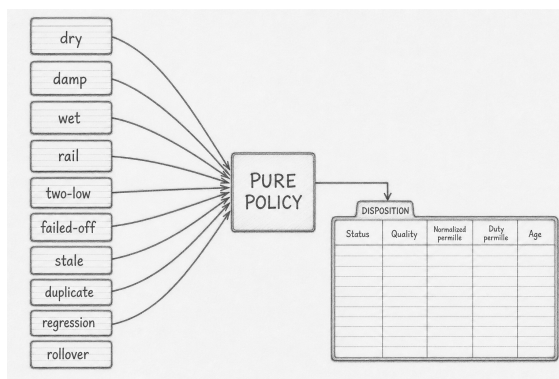
Sequence zero is always invalid. Accepting `UINT32_MAX` latches sequence exhaustion: every later update, including a duplicate, returns `CapacityExceeded` atomically. Only explicit reset clears the latch and begins a new volatile sequence epoch. This is bounded ordering behavior, not durable producer identity across a restart.

8 Replay the policy without hardware

The Mega sketch constructs copied samples and writes observations to a named result cell. It does not read a pin, energize a probe, acquire a timer, or claim a visible physical result. The HTML lesson is the hardware-independent route to the same learning outcome.

```
ResistiveProbeObservationPolicy probe(config);
probe.initialize();

sample.sequence += 1U;
sample.observdAt = now;
const Status updateStatus = probe.update(now, sample);
const ResistiveProbeObservation result = probe.snapshot();
```



Alternative text: a pencil replay ledger feeds dry, damp, wet, rail, two-low, failed-off, stale, duplicate, regression, and rollover sample cards into one policy and records status, quality, normalized value, duty, and age in named result cells.

The deterministic proof covers both calibration slopes; ADC endpoints; references and thresholds below, at, and above boundaries; dry-to-wet and contamination-drift ramps; open, rail, stuck, incomplete-discharge, backfeed, duty, stale, producer-fault, duplicate, regression, rollover, reset, and byte-stable replay traces.

The canonical replay also records the reset snapshot and initialization status, then admits a fresh sequence after reset. Reset clears the observation and ordering history while preserving the validated configuration and initialized lifecycle; a repeated `initialize()` is idempotent. The named cells establish only that volatile policy transition.

9 Inspect evidence before making a claim

Gate	Pass evidence	A pass still does not establish
resource acquisition	zero pins, ADC channels, timers, buses, endpoints, or supplies	powered circuit availability
safe state	pure reset and rejected updates preserve bounded policy state	physical excitation removal
classification	exact precedence and golden normalized values	liquid identity or depth
time/order	age, duplicate, regression, and rollover traces	sampling cadence between records
replay	byte-identical result ledger	bench repeatability

The canonical measurements are 5,662 bytes of flash and 624 bytes of static SRAM for the ordinary Mega replay, and 3,566 bytes of flash plus 169 bytes of static SRAM for the isolated no-LTO probe. The conservative synchronous stack is 123 bytes; the policy object is 69 bytes; and caller-owned input and output buffers are 27 and 37 bytes, or 64 bytes together. After the specified 128-byte ISR reserve, 7,772 bytes of Mega SRAM remain after static SRAM and the synchronous stack.

The ordinary sketch and isolated no-LTO probe are distinct measurement boundaries; their flash and static-SRAM figures are not one aggregate. The gate also checks ABI/ownership, retained-return stack edges, and the toolchain fingerprint. These compiler results do not substitute for electrical or physical evidence.

E1a remains open. Before any powered adapter, wiring, schematic, or support claim, qualify one exact probe and circuit: switched excitation, ADC reference and source impedance, off/backfeed behavior, current, duty waveform over the complete campaign, corrosion, spill containment, drying, calibration stability, and a non-Serial physical observation path.

10 Completion record

Record the exact evidence you inspected.

Check	Result or evidence
Both calibration slopes and threshold neighbors	
Excitation-off, discharge, on-time, and duty precedence	
Saturation and exact two-low disconnected predicate	
Fresh/stale boundary, duplicate, regression, and rollover	
Reset, replay determinism, and named result cell	
Canonical numeric resource report and all hard gates	
Zero-hardware E0 boundary and open E1a work	

Completion statement: I can distinguish copied acquisition evidence from physical qualification, apply structural and quality precedence, normalize either calibration slope, explain the narrow disconnected predicate and per-cycle duty limit, and identify every powered claim that remains open.

Learner: _____ Reviewer: _____ Date: _____

This print edition is untagged. Its text is searchable and copyable, and the HTML lesson provides the hardware-independent alternative. No PDF/UA, WCAG, or other PDF accessibility-conformance claim is made.